American Computer Science League

2022-2023 ● Contest 2: Short Problems Solutions ● Senior Division

1. Prefix/Infix/Postfix Notation

+ * 3 − 4 / 6 2 ^ + − 2 * 2 2 6 2 = + * 3 − 4 (/ 6 2) ^ + − 2 (* 2 2) 6 2
= + * 3 − 4 (6 / 2) ^ + − 2 (2 * 2) 6 2
= + * 3 (− 4 3) ^ + (− 2 4) 6 2
= + * 3 (4 − 3) ^ + (2 − 4) 6 2
= + (* 3 1) ^ (+ (-2) 6) 2
= + (3 * 1) ^ ((-2) + 6) 2
= + 3 (^ 4 2)
= + 3 (4 ^ 2)
= (+ 3 16)
= 3 + 16
= 19

B. 19

2. Prefix/Infix/Postfix Notation

$\sqrt{1 + \sqrt{1 + \sqrt{1 + x}}}$

$= \sqrt{1 + \sqrt{1 + ((1 + x) \uparrow (1 / 2))}}$

$= (1 + (1 + ((1 + x) \uparrow (1 / 2))) \uparrow (1 / 2)) \uparrow (1 / 2)$
$= (1 + (1 + ((1\ x\ +) \uparrow (1\ 2\ /))) \uparrow (1\ 2\ /)) \uparrow (1\ 2\ /)$
$= (1 + (1 + (1\ x + 1\ 2\ / \uparrow)) \uparrow (1\ 2\ /)) \uparrow (1\ 2\ /)$
$= (1 + (1\ 1\ x + 1\ 2\ / \uparrow + 1\ 2\ / \uparrow\ +)) (1\ 2\ /) \uparrow$
$= (1\ 1\ 1\ x + 1\ 2\ / \uparrow + 1\ 2\ / \uparrow\ +) (1\ 2\ /) \uparrow$
$= 1\ 1\ 1\ x + 1\ 2\ / \uparrow + 1\ 2\ / \uparrow + 1\ 2\ / \uparrow$

C. 111x+12/^+12/^+12/^

3. Bit-String Flicking

(RCIRC-2 (LSHIFT-1 ((RSHIFT-1 01110) AND (LCIRC-2 10011)
OR (NOT 01101))))
= (RCIRC-2 (LSHIFT-1 (00111 AND 01110 OR 10010)))
= (RCIRC-2 (LSHIFT-1 ((00111 AND 01110) OR 10010)))
= (RCIRC-2 (LSHIFT-1 (00110 OR 10010)))
= (RCIRC-2 (LSHIFT-1 10110))
= (RCIRC-2 01100)
= 00011

D. 00011

4. Bit-String Flicking

Let X = abcde

LHS = (LCIRC-2 X) OR (RSHIFT-2 01101) AND (NOT 11011)
= (LCIRC-2 abcde) OR (RSHIFT-2 01101) AND (NOT 11011)
= cdeab OR (00011 AND 00100)
= cdeab OR 00000
= cdeab
RHS = (LSHIFT-2 (RCIRC-2 10110))
= (LSHIFT-2 10101)
= 10100

LHS = RHS ⇒ cdeab = 10100 ⇒ c = 1, d = 0, e = 1, a = 0, b = 0
Therefore X = 00101

A. 00101

5. LISP

(SETQ M '(1 2 (5 1) (3 (4 5) 6) (2 (4))))
(CDR (CAR (CDR (REVERSE (CDR (CDR M))))))
= (CDR (CAR (CDR (REVERSE (CDR
(CDR '(1 2 (5 1) (3 (4 5) 6) (2 (4)))))))))
= (CDR (CAR (CDR (REVERSE (CDR '(2 (5 1) (3 (4 5) 6) (2 (4))))))))
= (CDR (CAR (CDR (REVERSE '((5 1) (3 (4 5) 6) (2 (4)))))))
= (CDR (CAR (CDR '((2 (4)) (3 (4 5) 6) (5 1)))))
= (CDR (CAR '((3 (4 5) 6) (5 1))))
= (CDR '(3 (4 5) 6))
= ((4 5) 6)

C. ((4 5) 6)